

SIEMENS EDA

Algorithmic C (AC) Communication **Release Notes**

Software Version v3.4.5
September 2022



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Release 3.4.5

This release is the first open-source release of the `ac_comm` library. The following table lists the communication blocks available in this release.

Description	Class Name(s)	Header File(s)
LTE MIMO Layer Mapper and Precoder	<code>ac_lte_map_precode</code> <i>Sub-blocks:</i> <code>ac_lte_map</code> <code>ac_lte_precode</code>	<code>ac_lte_map_precode.h</code>
Bit-to-Symbol Mapping	<code>ac_bit2symbol</code>	<code>ac_bit2symb_mapping.h</code>
LTE Rate 1/3 Tail Biting Convolutional Encoder	<code>ac_conv_encoder</code>	<code>ac_conv_encoder.h</code>
Denoising Filter	<code>ac_denoise_filter</code>	<code>ac_denoise_filter.h</code>
Advanced Encryption Standard	<code>ac_aes</code>	<code>ac_aes.h</code>
LDPC Encoder/Decoder	<code>ac_ldpc_5g_decoder</code> <code>ac_ldpc_5g_encoder</code>	<code>ac_ldpc_5g_decoder.h</code> <code>ac_ldpc_5g_encoder.h</code>
LTE Rate Matching	<code>ac_ratematch</code>	<code>ac_ratematch.h</code>
LTE Rate 1/3 Turbo Encoder	<code>ac_turbo_encoder</code>	<code>ac_turbo_encoder.h</code>

Supported Compilers

In order to use and run the *ac_comm* library, it is recommended that the user use the g++ compiler provided in GCC (Gnu Compiler Collective) version 4.8.5 or later, with the compiler standard being C++11 or later to support the usage of static asserts in the source code.